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Sociology 526
**Research Methods for
Social Network Analysis**

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“A collection of human beings does not become a society because each of them has an objectively determined or subjectively impelling life-content. It becomes a society only when the vitality of these contents attains the form of reciprocal influence; only when one individual has an effect, immediate or mediate, upon another, is mere spatial aggregation or temporal succession transformed into society. If, therefore, there is to be a science whose subject matter is society and nothing else, it must exclusively investigate these interactions, these kinds and forms of sociation.”

—Georg Simmel, “The Problem of Sociology” (1908)

Course Description

Whereas much sociology is focused on the analysis of *variables* (such as education and income) abstracted from observable relations among individuals and institutions, social network analysis is the study of structures of social relations. This seminar treats qualitative and quantitative methods for social network research, emphasizing a “how-to” approach to analyzing existing datasets or those collected by seminar participants in their own research.

Prerequisites

There are no formal prerequisites. As Degenne and Forsé (1999) write* (p. 12), “in some studies methodology goes straight to the Appendix, but not in network analysis. Most network analysts put methods at the heart of the analysis.” *Of course* I do not assume that you know about semigroup algebras, eigenvectors, or geodesic distances. My job is to get you comfortable with concepts such as these, by making them as concrete and research-relevant as possible. I am confident I can do this, but the “prerequisite” is motivation on your part to be interested in the practicalities of network analysis and its possible relevance to your research interests. This is a methods course with an attitude (see, e.g., the quotation above). It will help greatly if you have read widely in the social sciences, have conducted your own research or thought about doing so, and have some basic computer and internet skills.

Computer programs: In order to learn how to run and use computer programs for network analysis, I will introduce you to two main programs (UCINET and PAJEK) and, more briefly, to a programming language (MATLAB) in which I have written programs not

* Introducing Social Networks ([1994] 1999). A list of abbreviated titles appears on the course web page, under “Availability of Books.”

available through standard sources (and I will perhaps point you to some other programs as we move along). UCINET is a user-friendly network analysis suite of programs: you just “point and click.” It is the most widely-used software for network analysis. PAJEK is a moderately-friendly suite of programs emphasizing visualization of networks and “generalized blockmodeling.” Good news! All (with the partial exception of UCINET) are available to you at no charge. You can download UCINET and PAJEK, and you already have access to MATLAB on the mainframe if you have a computer account at this university. To learn more or to download these programs (and manuals and datasets), see the course web page. Also, and importantly, UCINET is available to everyone on many computers in the SBS computer lab (Social Sciences Bldg. room 224) and (to sociology grad students) in the DASL lab.

Books: We’ll make extensive use of the Wasserman and Faust text. In addition, this year for the first time I’m incorporating the 2005 text on PAJEK by de Nooey et al. Whereas Wasserman and Faust is something of an all-encompassing encyclopedia, the PAJEK text is more idiosyncratic and more oriented toward substantive research questions, and it has great sets of assignments and exercises (including answers!). You may wish to look through the reading list or the books themselves (they will be available on 24-hour reserve in the Main Library as well as in the bookstore) before deciding which other books to purchase.

Readings: Most will be available on the course web site. Take a look!

Course web page: I will make extensive use of a course web site, D2L, sponsored by the University of Arizona Learning Technology Center. **You will find it helpful to “click” often on this site, probably doing so before every class meeting!** The web address (url) is:

<http://d2l.arizona.edu/>

Please “bookmark” this location on your home computer, for easy future reference. Once at the above location, use the “NetID Login” option. If you have enrolled in the course, you should be recognized. (Otherwise, see me.)

Requirements

1. Full participation in a seminar of this type is desirable, and needs to be based on thorough preparation for each class. (*Read* the material and *think* about it before each class. Please note that *the reading list is not as long as it seems*—see the Note at the top of p. 4). Some of the readings are super-technical, so—not uncommonly—you won’t understand some readings. My goal however is to enable you to understand, criticize, and apply the major approaches we’ll be learning – through a combination of readings, class lectures, class discussion, handouts, using computer programs, and focusing on examples. Participation: 20% of final seminar grade.

2. I will often assign homework, and occasionally collect it. All homework is **ungraded**. These assignments will usually involve repeating some analysis that Wasserman and Faust illustrate in their textbook. The aim is to empower you as a network researcher who can conduct analyses and understand what you're doing. Simply completing the homework will count in determining your final grade (20%).
3. There will be a midterm exercise. I might give you a published article and the network data on which it is based, and ask you to discuss/criticize/extend the author's analysis by means of playing around with the same data, by using the programs we will have discussed in class. This is an *exercise*, not a magnum opus. (20% of final grade.)
4. A term paper, which will probably take one of three forms. Probably the most common form will be a data analysis paper, either analysis of data you've collected or a re-analysis of data made available through the course. But two other forms are also possible for the term paper: a conventional library-research term paper ("Social Network Imagery in the Novels of Balzac" and "Marketing Research [or the Sociology of Law, etc.] and Social Networks" are two among a very wide set of possible topics), or a critical essay ("What's Wrong with Network Analysis" or "Bringing Together Social Networks, Rational Choice, Ethnomethodology, and Marxist Post-Structuralism" are possibilities). The paper will be due one week after the last class meeting. *Please talk with me as the semester goes along about your ideas for the paper, outlines, and your progress in writing the paper.* 40% of final course grade. I will place an interesting collection of books on reserve in the Main Library that may stimulate interest in some paper topics; see also the course web site.

Reading List

Please Note: (1) I do not assume that we will read all items listed under each class.

Some items are listed for the sake of (increased) completeness, or for participants who have special interests in certain topics. As we go along the distinction between assigned and optional readings will be made clear. **(2)** Citations briefly listed are provided in full on the course web site (under “Administrative Stuff / Books on Reserve”). **(3)** Many, in fact the great majority, of the course readings (other than those in the Wasserman-Faust and de Nooey et al. texts) are available at the D2L course web site. **(4)** There are more topics listed than there are class meetings. A few topics will take more than one class meeting to cover. A few will need to be left out; if you have views on what to retain and what to cut, please let me know.

0. Introduction to the Course and its Objectives

These readings and activities are **not required** at this point, but are listed in case you may find them to be helpful. All (except Linton Freeman’s 2004 book) may be found under READINGS on the course web page. Freeman 2004 is available at the Main Library.

Barry Wellman’s powerpoint slides introducing network analysis: you can find these on the course web page, under READINGS.

David Knoke’s powerpoint slides introducing his network analysis course.

Ron’s slides from the first class lecture, also on the course web page.

Linton C. Freeman, The Development of Social Network Analysis: A Study in the Sociology of Science, Vancouver: BC Press, 2004. Esp. Ch. 1.

Joseph Galaskiewicz and Stanley Wasserman, “Social Network Analysis: Concepts, Methodology, and Directions for the 1990s.” Sociological Methods & Research **22** (1993): 3-22.

R.L. Breiger, pp. 505-07 of “The Analysis of Social Networks,” in Melissa Hardy and Alan Bryman (eds.), Handbook of Data Analysis (London: Sage, 2004).

PART I. The Discovery of Social Networks

1. How do people acquire or construct useful information?

Clinton B. DeSoto, “Learning a Social Structure” [1960], reprinted pp. 99-103 in Samuel Leinhardt (ed.), Social Networks: A Developing Paradigm (New York: Academic Press, 1977).

James Coleman, Elihu Katz, and Herbert Menzel, "The Diffusion of an Innovation among Physicians" [1957], reprinted pp. 107-124 in Leinhardt, (see above).

2. When can your “weak” connections to other people be “strong”?

Mark S. Granovetter, "The Strength of Weak Ties," American Journal of Sociology **78** (1973): 1360-1380.

Pp. **18-20** and **25-30** of Ronald S. Burt, Structural Holes: The Social Structure of Competition (Harvard University Press, 1992).

Pp. **127-28** in R.L. Breiger, "Social Structure and the Phenomenology of Attainment," Annual Review of Sociology **21** (1995): 115-36.

Elihu Katz and Paul F. Lazarsfeld, "The Rediscovery of the Primary Group: Case Histories of an Intervening Variable," pp. 31-42 in Katz & Lazarsfeld, Personal Influence: The Part Played by People in the Flow of Mass Communications (New York, Free Press, 1955).

3. Network Data Collection

Peter V. Marsden, "Network Data and Measurement," Annual Review of Sociology **16** (1990): 435-63.

Peter V. Marsden, "Recent Developments in Network Measurement," in P.J. Carrington, J. Scott, and S. Wasserman (eds.), Models and Methods in Social Network Analysis (Cambridge University Press, 2005), 8-30.

Pp. **627-42** of Daniel A. McFarland, "Student Resistance: How the Formal and Informal Organization of Classrooms Facilitate Everyday Forms of Student Defiance," American Journal of Sociology **107** (2001): 612-78.; much of the rest of the article is studied in Part VII of the course.

Esp. pp. **51-53** in Rob Cross, Wayne Baker, and Andrew Parker, "What Creates Energy in Organizations?" MIT Sloan Management Review **44** (Summer 2003): 51-56.

Mario Diani, pp. **175-85** of "Network Analysis," in Bert Klandermans and Suzanne Staggenborg (eds.), Methods of Social Movement Research (University of Minnesota Press, 2002, pp. 173-200).

Elisabeth S. Clemens and Martin D. Hughes, pp. **209-14** of "Recovering Past Protest: Historical Research on Social Movements," Klandermans & Staggenborg (*ibid.*, pp. 201-30).

Paul Lichterman, pp. **138-39** of "Seeing Structure Happen: Theory-Driven Participant Observation," in Klandermans & Staggenborg (*ibid.*, pp. 118-45).

Esp. pp. **311-20** in Katherine Faust, B. Entwisle, R.R. Rindfuss, S.J. Walsh, & Y. Sawangdee, "Spatial Arrangement of Social and Economic Networks Among Villages in Nang Rong District, Thailand," Social Networks **21** (1999): 311-37.

David Knoke and James H. Kuklinski on “Basic Concepts” and “Data Collection,” pp. 9-35 of Network Analysis.

R.L. Breiger, pp. **507-10** in “Social Network Analysis.” (handbook chapter).

Wasserman & Faust, “Data Sets Found in These Pages,” pp. 59-66.

4. Introduction to the Pajek program (1): Data format and visualization

de Nooey, Mrvar, and Batagelj, Exploratory Social Network Analysis with Pajek, ch. 1 (pp. 3-28).

5. Introduction to the Pajek program (2): Attributes and relations

de Nooey, Mrvar, and Batagelj, ch. 2 (pp. 29-57).

Part II. Some Structural Properties and Measures

6. Cohesive Subgroups

de Nooey et al., ch. 3 (pp. 61-83).

7. Centrality and Structure (and an introduction to the UCINET program)

Wasserman & Faust, Social Network Analysis, p. 110 (definition of “geodesic”) **and** pp. 177-192 (on three types of centrality measure); more generally, chapter 5.

Knoke & Kuklinski, pp. 53-56.

Linton C. Freeman, “A Set of Centrality Measures Based on Betweenness,” Sociometry **40** (1977): 35-41.

Pp. **1274-80** (see esp. the footnotes on p. 1278) in John F. Padgett and Christopher K. Ansell, “Robust Action and the Rise of the Medici, 1400-1434.” American Journal of Sociology **98** (1993): 1259-1319.

Arthur S. Alderson and Jason Beckfield, “Power and Position in the World City System.” American Journal of Sociology **109** (2004): 811-51.

Wayne E. Baker and Robert R. Faulkner, “The Social Organization of Conspiracy: Illegal Networks in the Heavy Electrical Equipment Industry,” American Sociological Review **58** (1993): 837-60.

8. Positive and Negative Eigenvector Centrality: Different measures are needed for distinctive models of networks

Val Burris, "The Academic Caste System: Prestige Hierarchies in PhD Exchange Networks." American Sociological Review **69** (2004): 239-264.

Naomi Rosenthal, Naomi, Roberta Karant, Michele Ethier, and Meryl Fingrutd, "Centrality Analysis for Historians." Historical Methods **20** (1987): 53-62.

Phillip Bonacich, "Power and Centrality: A Family of Measures." American Journal of Sociology **92** (1987): 1170-1182.

9. Brokers and Bridges

de Nooy et al., ch. 7 (pp. 138-160)

Judd H. Michael, "Labor Dispute Reconciliation in a Forest Products Manufacturing Facility." Forest Products Journal **47** (11/12, Nov/Dec), 1997, pp. 41-45.

James Moody and Douglas R. White, "Structural Cohesion and Embeddedness: A Hierarchical Concept of Social Groups." American Sociological Review **68** (2003): 103-127.

10. Structural Holes and Social Capital

Ronald S. Burt, pp. 8-49 in Structural Holes, 1992.

Stephen P. Borgatti, "Structural Holes: Unpacking Burt's Redundancy Measures," in Connections: Official Journal of the International Network for Social Network Analysis **20** (Summer 1997): 35-38.

Degenne and Forsé, "Social Capital," pp. 107-131 (but esp. **117-19**) in Introducing Social Networks.

11. Clustering and Small Worlds – and Research Applications

Pp. 41-54 ("Small Worlds") in Albert-László Barabási, Linked: The New Science of Networks (2002).

Duncan J. Watts and Steven H. Strogatz, "Collective Dynamics of 'Small-World' Networks." Nature 393 (1998): 440-42.

Phillip Bonacich, [book review] "Invasion of the Physicists." Social Networks **26** (2004): 285-8.

Mark Newman, "The Structure of Scientific Collaboration Networks." Proceedings of the National Academy of Sciences **98** (2001): 404-9.

Valdis E. Krebs, "Mapping Networks of Terrorist Cells." Connections 24 (December 2001): 43-52.

Valdis E. Krebs, "Uncloaking Terrorist Networks." First Monday 7 (April 2002), e-journal on the web (see course page WEBLIOGRAPHY).

Duncan J. Watts. "Networks, Dynamics, and the Small-World Phenomenon." American Journal of Sociology 13 (1999): 493-527.

12. Centrality, Holes, Personality, and Good Ideas

Ajay Mehra, Martin Kilduff, and Daniel J. Brass, "The Social Networks of High and Low Self-Monitors: Implications for Workplace Performance." Administrative Science Quarterly 46 (2001): 121-46.

Ronald S. Burt, "Structural Holes and Good Ideas." American Journal of Sociology 110 (2004): 349-99.

R.S. Burt, J.E. Jannotta, and J.T. Maloney, "Personality Correlates of Structural Holes." Social Networks 20 (1998): 63-82.

Yuval Kalish and Garry Robins, "Psychological predispositions and network structure: The relationship between individual predispositions, structural holes and network closure." Social Networks 28 (2006): 56-84.

13. Duality and Affiliation Networks

Georg Simmel, "How is Society Possible?" (pp. 6-22), "The Problem of Sociology" (pp. 23-35), and possibly "Group Expansion and the Development of Individuality" (pp. 251-93) in Donald Levine (ed.), Georg Simmel on Individuality and Social Forms.

R.L. Breiger, "The Duality of Persons and Groups," Social Forces 53 (1974): 181-90 (reprinted in Wellman and Berkowitz, Social Structures, 83-98).

Wasserman & Faust, pp. 291-326.

Peter S. Bearman, "The Social Structure of *Suicide*," Sociological Forum 6 (1991): 501-524.

Phillip Bonacich, Annie Cody Holdren, and Michael Johnston, "Hyper-Edges and Multidimensional Centrality." Social Networks 26 (2004): 189-203.

Thomas J. Fararo and Patrick Doreian, "Tripartite Structural Analysis: Generalizing the Breiger-Wilson Formalism." Social Networks 6 (1984): 141-175.

14. Lattice Analysis and Correspondence Analysis

Wasserman & Faust, pp. 326-42.

R.L. Breiger, pp. **102-04** ["How to construct and interpret a Galois lattice"] and **99-102** [on correspondence analysis] in "A Tool Kit for Practice Theory." Poetics **27** (2000): 91-115.

Linton C. Freeman, "Cliques, Galois Lattices, and the Structure of Human Social Groups." Social Networks **18** (1996): 173-187.

Katherine Faust, "Centrality in Affiliation Networks," Social Networks **19** (1997): 157-191, esp. 157-66 and 169-72.

15. An Ecology of Affiliation

J. Miller McPherson, "An Ecology of Affiliation." American Sociological Review **48** (1983): 519-532.

Miller McPherson, Lynn Smith-Lovin, and James M. Cook, "Birds of a Feather: Homophily in Social Networks." Annual Review of Sociology **27** (2001): 415-44.

Noah Mark, "Birds of a Feather Sing Together." Social Forces **77** (1998): 453-485

Part III. Structural Equivalence: Concrete Roles and Positions

16. Structural Equivalence Pure and Dirty: An introduction

Wasserman & Faust on structural equivalence, pp. 354-393.

Knoke & Kuklinski on structural equivalence, pp. (36-37 and) 59-77.

Chun-Houh Chen, "Generalized Association Plots: Information Visualization via Iteratively Generated Correlation Matrices." Statistica Sinica **12** (2002): 7-29.

Pp. 12-14 and 17-24 in John M. Sharp, Eui Hang Shin, and LeRoy F. Smith, "A Network Analysis of Departmental Prestige Based on Origins of Faculty Degrees." Behavioral Science **27** (1982): 12-25.

17. Blockmodels of Roles and Positions

Harrison C. White, Scott A. Boorman, R.L. Breiger, "Social Structure from Multiple Networks: I. Blockmodels of Roles and Positions." American Journal of Sociology **81** (1976): 730-780.

Wasserman & Faust on blockmodels, pp. 394-424. [Also, pp. 679-88, on goodness of fit]

R.L. Breiger, pp. 510-14 in "Social Network Analysis." (2004 handbook chapter).

Special issue on blockmodels, Social Networks **14** (1992): 1-186.

18. Applications

Nancy J. Herman, "Conflict in the Church: A Social Network Analysis of an Anglican Congregation." Journal for the Scientific Study of Religion **23** (1984): 60-74.

Paul DiMaggio, "Structural Analysis of Organizational Fields: A Blockmodel Approach," pp. 335-370 in Barry M. Staw and L. L. Cummings (eds.), Research in Organizational Behavior **8** (1986). Westport, CT: JAI Press.

Anheier, Helmut K., Jürgen Gerhards, and Frank P. Romo, "Forms of Capital and Social Structure in Cultural Fields: Examination of Bourdieu's Social Topography." American Journal of Sociology **100** (January 1995): 859-903.

Robert R. Faulkner, Music on Demand: Composers and Careers in the Hollywood Film Industry (1983), pp. 185-99 and 218-21.

Katherine Giuffre, "Sandpiles of Opportunity: Success in the Art World." Social Forces **77** (1999): 815-32.

Mario Diani, "Britain Re-Creates the Social Movement: Contentious (and Not-so-contentious) Networks in Glasgow," unpublished (October 2002).

Michael L. Gerlach, "The Japanese Corporate Network: A Blockmodel Analysis," Administrative Science Quarterly **37** (March 1992): 105-139.

Part IV. Some Statistical Models for Networks

19. Holland and Leinhardt's " p_1 " model

[For background:] R.L. Breiger, p. 514 "Social Network Analysis." (2004 handbook chapter).

Wasserman and Faust, pp. 605-35 [This chapter was written by Dawn Iacobucci].

R.L. Breiger, "Comment [on Holland-Leinhardt 1981]." Journal of the American Statistical Association **76** (1981): 51-53.

Joseph Galaskiewicz and Stanley Wasserman, "Mimetic and Normative Processes within an Organizational Field: An Empirical Test." Administrative Science Quarterly **34** (1989): 454-479.

20. Stochastic Blockmodels: Generalizations of p_1

Yuchung J. Wang and George Y. Wong, "Stochastic Blockmodels for Directed Graphs." Journal of the American Statistical Association **82** (1987): 8-19.

Wasserman and Faust, pp. 692-706.

Paul W. Holland, Katherine B. Laskey, and Samuel Leinhardt, "Stochastic Blockmodels: Some First Steps." Social Networks 5 (1983): 109-137.

Tom A.B. Snijders and Krzysztof Nowicki, "Manual for BLOCKS version 1.51" (2001), especially pp. 4-7 and 12-26 [available as a download from the course WEBLIOGRAPHY under "program acquisition –StOCNET"].

Nowicki & Snijders, "Estimation and Prediction for Stochastic Blockstructures." Journal of the American Statistical Association 96 (2001): 1077-87.

21. The p* Model of Wasserman, Pattison, and Robbins

Stanley Wasserman and Philippa E. Pattison, "Logit models and logistic regressions for social networks: I. An introduction to Markov graphs and p*." Psychometrika 60 (1996): 401-426.

Breiger, pp. 514-17 of "Social Network Analysis," Handbook chapter (2002).

22. Application of "p*": Exchanges within an Organization

Emmanuel Lazega and P.E. Pattison, "Multiplexity, Generalized Exchange and Cooperation in Organizations: A Case Study." Social Networks 21 (1999): 67-90.

23. New progress in estimating random graph (p*) models

Work of Gary Robins et al., including programs – to be added.

G. Robins, T. Snijders, P. Wang, M. Handcock, P. Pattison, "Recent Developments in Exponential Random Graph (p*) Models for Social Networks." Social Networks, to appear.

24. Correlation and Regression for Networks: QAP models

N. Mantel, "The Detection of Disease Clustering and a Generalized Regression Approach." Cancer Research 27 (1967): 209-220.

David Krackhardt, "Predicting with networks: Nonparametric multiple regression analysis of dyadic data." Social Networks 10 (1988): 359-381.

Val Burris, "Interlocking Directorates and Political Cohesion among Corporate Elites." American Journal of Sociology 111 (2005): 249-283.

Part V. Generalized Equivalences: Abstract Roles and Positions

25. Preliminaries: Balance and Clusterability

James A. Davis, "Clustering and Structural Balance in Graphs," pp. 27-34 in Leinhardt, ed., Social Networks.

Wasserman & Faust on structural balance, ch. 6, pp. 220-48

26. Relational Algebras

R.L. Breiger and Philippa E. Pattison, "The Joint Role Structure of Two Communities' Elites," pp. 326-36 in Burt and Minor, Applied Network Analysis.

Scott A. Boorman and Harrison C. White, "Social Structure from Multiple Networks: II. Role Structures." American Journal of Sociology **81** (1976): 1384-1446.

Wasserman & Faust, pp. 425-460.

27. Automorphic and Regular Equivalence

Stephen P. Borgatti and Martin G. Everett, "Notions of Position in Social Network Analysis," pp. 1-35 in P.V. Marsden (ed.), Sociological Methodology 1992. Washington, DC: American Sociological Association.

Breiger, handout.

Wasserman & Faust on general equivalence, pp. 461-83.

Ronan Van Rossem, "The World System Paradigm as General Theory of Development: A Cross-National Test." American Sociological Review **61** (1996): 508-27.

Vladimir Batagelj, Patrick Doreian, and Anuška Ferligoj, "An Optimizational Approach to Regular Equivalence." Social Networks **14** (1992): 121-135.

Katherine Faust, "Comparison of Methods for Positional Analysis: Structural and General Equivalences." Social Networks **10** (1988): 313-341.

28. Generalized Blockmodeling

Patrick Doreian, "An Intuitive Introduction to Blockmodeling with Examples." Bulletin de Methodologie Sociologique **61** (January 1999).

de Nooey et al., "Blockmodels." Ch. 12 (pp. 259-291).

Patrick Doreian, Vladimir Batagelj, and Anuška Ferligoj, Generalized Blockmodeling (Cambridge University Press, 2005).

Peter Marsden, review of Doreian et al. (2005), in Social Networks **28** (2006): 275-282.

29. Local Roles and Ego Algebras

Michael J. Mandel, "Local Roles and Social Networks." American Sociological Review **48** (1983): 376-386.

Wasserman & Faust on local roles and ego algebras, 483-502.

R.L. Breiger and P.E. Pattison, "Cumulated Social Roles: The Duality of Persons and Their Algebras," Social Networks **8** (1986): 215-256.

Part VI. Some Further Topics

30. Diffusion

de Nooey et al., ch. 8 (pp. 161-183).

31. Small Worlds, Cluster Coefficients, and Bipartite Networks

M.E.J. Newman, S.H. Strogatz, and D.J. Watts, "Random graphs with arbitrary degree distributions and their applications," Physical Review E **64** (2001), 1-17, esp. pp. 1-2, 4-5, 11-16.

Brian Uzzi and Jarrett Spiro, "Collaboration and Creativity: The Small World Problem." American Journal of Sociology **111** (2005): 447-504.

32. Social Logics from Networks of Count Data

R.L. Breiger and John W. Mohr, "Institutional Logics from the Aggregation of Organizational Networks: Operational Procedures for the Analysis of Counted Data." Computational & Mathematical Organization Theory **10** (2004): 17-43.

Yanjie Bian, Ronald Breiger, Deborah Davis, and Joseph Galaskiewicz, "Occupation, Class, and Social Networks in Urban China." Social Forces **83** (2005): 1443-1468.

33. Role Equivalence via Triad Profiles

Ronald S. Burt, "Detecting Role Equivalence." Social Networks **12** (1990): 83-97.

Paul W. Holland and Samuel Leinhardt, "Local Structure in Social Networks." Sociological Methodology **1976** (ed. David R. Heise; Jossey-Bass, 1976), pp. 1-45.

Part VII. Network Analysis in Society

34. What makes networks effective or ineffective?

Pp. **620-27** and **662-68** in Daniel A. McFarland, “Student Resistance: How the Formal and Informal Organization of Classrooms Facilitate Everyday Forms of Student Defiance.” American Journal of Sociology **107** (2001): 612-78.

Jeffrey C. Johnson, Lawrence A. Palinkas, and James S. Boster, “Informal Social Roles and the Evolution and Stability of Social Networks,” pp. 121-32 in R.L. Breiger, K.M. Carley, and P.E. Pattison (eds.), Dynamic Social Network Modeling and Analysis: Workshop Summary and Papers. Washington, DC: The National Academies Press, 2003.

Elisa Jayne Bienenstock and Phillip Bonacich, “Balancing Efficiency and Vulnerability in Social Networks,” pp. 253-64 in Breiger et al. (eds), ibid.

35. Movement, Military, and Intelligence Applications

John Arquilla and David Ronfeldt, “The Advent of Netwar (Revisited),” pp. 1-25 in Arquilla and Ronfeldt, Networks and Netwars. Santa Monica, CA: RAND, 2001.

U.S. Army counterinsurgency manual (June 2006 draft), posted on the internet by the Federation of American Scientists, p. 3-21 and App. E on social network analysis.

Michael Hardt and Antonio Negri, Multitude: War and Democracy in the Age of Empire (New York: Penguin, 2004), section on “Asymmetry and Full-Spectrum Dominance,” pp. 51-62.

36. Ethical Dilemmas in Social Network Analysis

Social Networks special issue: **27** (2), 2005:

R.L. Breiger, “Ethical Dilemmas in Social Network Research: Introduction to Special Issue,” pp. 88-93.

Rebecca Goolsby, “Ethics and Defense Agency Funding: Some Considerations,” pp. 95-106.

Stephen P. Borgatti and José-Luis Molina, “Toward Ethical Guidelines for Social Network Research in Organizations,” pp. 107-117.

Alden S. Klov Dahl, “Social Network Research and Human Subjects Protection: Toward More Effective Infectious Disease Control,” pp. 119-137.

Charles Kadushin, “Who Benefits from Network Analysis: Ethics of Social Network Research,” pp. 139-153.

37. Network studies and Questions of Meaning

Clyde Mitchell, “Situational Analysis and Network Analysis,” Connections **17** (1994): 16-22.

Harrison White, pp. 67-70, 81-83, and 89 of Identity and Control (Princeton University Press, 1992).

Gary Alan Fine and Sherryl Kleinman, "Network and Meaning: An Interactionist Approach to Structure," Symbolic Interaction 6 (1983): 97-110.

Ann Mische, "Cross-Talk in Movements: Reconceiving the Culture-Network Link" (conference paper, June 2000; not the published version that appears as pp. 258-80 in Mario Diani and Doug McAdam, eds., Social Movements and Networks: Relational Approaches to Social Action, Oxford University Press, 2003).